

## Chapter 13

# Non-Conventional Sources of Energy

## Class 10 - Total Geography Morning Star

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### Choose the correct option

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#### Question 1

Most of the non-conventional sources of energy are ..... .

1. exhaustible
2. inexhaustible
3. polluting
4. expensive

#### *Answer*

inexhaustible

#### Question 2

India has higher amount of solar energy because of India's location on the ..... .

1. Tropic of Cancer
2. Tropic of Capricorn
3. Poles
4. Equator

#### *Answer*

Tropic of Cancer

#### Question 3

Most parts of India have ..... clear sunny days in a year.

1. 365
2. 350

3. 300
4. 200

***Answer***

300

**Question 4**

Solar cells are known as ..... cells.

1. Super
2. Photonic
3. Photovoltaic
4. Solaratic

***Answer***

Photovoltaic

**Question 5**

Solar energy is used to ..... water in regions with limited clean water supplies.

1. boil
2. freeze
3. condense
4. distill

***Answer***

boil

**Question 6**

Wind power accounts for ..... percent of India's total installed power generation capacity.

1. 10
2. 20
3. 25
4. 40

***Answer***

**Question 7**

Tidal energy can be harnessed at ..... .

1. Gulf of Kutch
2. Sundarbans
3. Andaman and Nicobar Islands
4. All of the above.

**Answer**

All of the above

**Question 8**

Geothermal energy is independent of ..... and season.

1. time
2. atmosphere
3. weather
4. winds

**Answer**

weather

**Question 9**

The process of splitting of radioactive substances is called ..... .

1. nuclear fission
2. nuclear fusion
3. nuclear reaction
4. nuclear bombing

**Answer**

nuclear fission

**Question 10**

Nuclear power is the ..... largest source of electricity in India.

1. second
2. fourth
3. fifth
4. sixth

**Answer**

fourth

## Answer the following questions

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### Question 1(a)

What are non-conventional sources of energy?

**Answer**

Non-Conventional sources of energy are those which have been developed in the recent past as an alternative to conventional sources of energy. These include solar energy, wind energy, tidal energy, geothermal energy, nuclear energy and biogas.

### Question 1(b)

State two differences between conventional and non-conventional sources of energy.

**Answer**

| Conventional sources of energy                                                                                                       | Non-Conventional sources of energy                                                                             |
|--------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Conventional sources of energy are those which have been used by man for a long time and are still being tapped and used abundantly. | Non-conventional sources of energy are those developed in the recent past as an alternative sources of energy. |
| Except for hydel energy, these sources of energy are non-renewable.                                                                  | Most of these sources are renewable.                                                                           |

### Question 1(c)

Give a reason for each of the following:

- (i) Non-conventional sources of energy are called the resources of the future.
- (ii) India is in an advantageous position in terms of solar energy.

(iii) Harvesting of solar energy does not require much maintenance.

**Answer**

(i) Non-conventional sources of energy are called the resources of the future as most of these sources of energy are renewable, non-polluting, less expensive to run and easy to maintain.

(ii) India is in an advantageous position to harness solar energy because of India's location on the Tropic of Cancer. Most parts of the country have 300 clear sunny days in a year and per hour per square km availability of solar energy between 5 to 7 kW.

(iii) Harvesting of solar energy does not require much maintenance as most reliable solar panel manufacturers give 20-25 years warranty. Also, as there are no moving parts, there is no wear and tear. So, after covering the initial cost of the solar energy system, it requires very little spending on maintenance and repair work.

**Question 1(d)**

How is India in an advantageous position to harness solar energy?

**Answer**

India is in an advantageous position to harness solar energy because of India's location on the Tropic of Cancer. Most parts of the country have 300 clear sunny days in a year and per hour per square km availability of solar energy between 5 to 7 kW.

**Question 2(a)**

State briefly how the Sun's energy can be used to generate power.

**Answer**

Sun's energy can be transformed into electrical energy with the help of solar cells. Solar cells are also known as photovoltaic cells. They are made up of thin wafers of semi conductor materials from silicon and gallium. When Sun's light falls on them, the Sun's energy gets converted into electricity.

**Question 2(b)**

Wind energy can generate additional income for the land owners.

**Answer**

Land owners can look for additional income by installing wind turbines on land that can even be used for agricultural purposes. The electricity generated by wind power can be used for domestic consumption and reduce monthly electricity bills, and the surplus power can be sold back to the

local grid which can result in more savings. Hence, wind energy can generate additional income for the land owners.

**Question 2(c)**

State any three advantages of using solar energy.

**Answer**

Three advantages of using solar energy are-

1. It is a renewable source of energy which can be harnessed in most of the areas of the world.
2. It can be used for diverse purposes like producing electricity in areas without access to the energy grid, to distill water in regions with limited clean water supplies and to power satellites in space.
3. Solar energy systems do not require a lot of maintenance.

**Question 2(d)**

How is wind energy utilised to produce electricity?

**Answer**

The wind energy is obtained by making use of windmills. The blades of the windmill rotate due to the force of the wind. The rotational motion of the blades drives a number of machines like water pumps, flour mills and electric generators. A number of windmills are installed in a definite pattern in clusters called wind farms. They generate a large amount of electricity.

**Question 3(a)**

What is tidal energy? Give an advantage of using tidal energy.

**Answer**

The rise and fall of water of the ocean produced by the attraction of the Moon and Sun produces a large amount of energy called Tidal energy.

An advantage of Tidal energy is that it can be produced even if the water moves at a low speed.

**Question 3(b)**

Give two advantages of using wind energy.

**Answer**

Two advantages of using wind energy are-

1. It is a renewable source of energy that can be produced again and again.
2. It is the cleanest form of renewable energy that produces no pollution.

**Question 3(c)**

What are wind farms? Where are they usually located?

**Answer**

A number of windmills are installed in a definite pattern in clusters called wind farms.

They are usually installed in coastal regions, open grasslands or hilly regions.

**Question 3(d)**

- (i) What is geothermal energy?
- (ii) How is geothermal energy used to produce electricity?

**Answer**

(i) Geothermal energy is the heat energy contained in the rock and fluid that fills the fractures and pores within the rock in the earth's crust.

(ii) Resources of geothermal energy can be generated by using the shallow steam, hot water and hot rock by drilling wells up to thousands of feet beneath the earth's surface. The extremely high temperatures in the deeper geothermal reservoirs are used for the generation of electricity. The high-pressure steam spins the turbine that rotates a generator and produces electricity.

Hot water is pumped from deep underground through a well under high pressure. When the water reaches the surface, the pressure is dropped, which causes the water to turn into steam. The steam spins the turbine, which is connected to a generator that produces electricity. The steam cools off in the cooling tower and condenses back to water. The cooled water is pumped back into the earth to begin the process again.

**Question 4(a)**

Name one area each where solar, wind, tidal and geothermal energy is generated in India.

**Answer**

| Energy            | Area                                  |
|-------------------|---------------------------------------|
| Solar energy      | Bhadla Solar Park, Jodhpur, Rajasthan |
| Wind energy       | Madurai in Tamil Nadu                 |
| Tidal energy      | Gulf of Kutch                         |
| Geothermal energy | Manikaran in Himachal Pradesh         |

#### Question 4(b)

State any two advantages of using geothermal energy.

#### Answer

Two advantages of using geothermal energy are-

1. It is easily accessible because of its extensive distribution.
2. It is independent of weather and season.

#### Question 4(c)

Name the process involved in generation of nuclear energy. State any three advantages of using nuclear power.

#### Answer

The process involved in generation of nuclear energy is known as **Nuclear Fission**.

Three advantages of using nuclear power are —

1. Nuclear energy reduces the requirement of the amount of energy generated from fossil fuels and therefore, reduces the use of fossil fuels and lowers Greenhouse Gas emissions.
2. Fission of one atom of uranium produces ten million times the energy released by burning of one atom of carbon. In this way, nuclear energy not only saves on raw materials but also in transport, handling and extraction of nuclear fuel. The cost of nuclear fuel is 20% of the cost of energy generated.
3. Nuclear energy is the most reliable source of energy as nuclear power plants can run 24 hours a day and seven days a week.

#### Question 4(d)



How is biogas beneficial for the farmers and householders?

**Answer**

Biogas is beneficial for the farmers in the form of energy and good quality manure. It is useful for the householders as the gas is used for cooking, lighting and pumping water from wells.

**Question 5(a)**

What is nuclear power?

**Answer**

Nuclear power is obtained from energy stored in the nuclei of atoms of naturally occurring radioactive elements like uranium, thorium and plutonium.

**Question 5(b)**

Name two minerals used for the generation of nuclear energy. Name two nuclear power plants in India.

**Answer**

Uranium and Thorium are the two minerals used for the generation of nuclear energy.

Two nuclear power plants in India are-

1. Kaiga in Karnataka
2. Kakrapar in Gujarat

**Question 5(c)**

What is biogas? How is biogas produced?

**Answer**

Biogas is composed of methane, carbon dioxide, hydrogen and hydrogen sulphide.

Biogas is produced by anaerobic degradation of animal and plant wastes in the presence of water. Anaerobic degradation means breaking down of organic matter by bacteria in the absence of oxygen. The fermentation of the waste products is carried out by bacteria, which produces hydrocarbon gas, predominantly methane and alcohol. The residue left behind in the tank is rich in nutrients and can be used as manure.

**Question 5(d)**

Explain how using nuclear energy in a controlled environment can be a boon for our country.

### ***Answer***

Using nuclear energy in a controlled environment can be a boon for our country for the following reasons-

1. Nuclear energy reduces the requirement of the amount of energy generated from fossil fuels and therefore, reduces the use of fossil fuels and lowers Greenhouse Gas emissions.
2. Fission of one atom of uranium produces ten million times the energy released by burning of one atom of carbon. In this way, nuclear energy not only saves on raw materials but also in transport, handling and extraction of nuclear fuel. The cost of nuclear fuel is 20% of the cost of energy generated.
3. Nuclear energy is the most reliable source of energy as nuclear power plants can run 24 hours a day and seven days a week. Besides, a nuclear plant needs to refuel every 18 to 24 months. This shows that nuclear fuels are less affected by fluctuating fuel prices than the more volatile industries like oil and natural gas.
4. When compared to the fossil fuel waste, the nuclear waste which occurs due to the production of nuclear power is small in quantity.

## **Thinking Skills**

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### **Question 1**

Which non-conventional source of energy would you prefer to use for running street lights, water pumps for irrigation and heating your house? Give reasons to support your answer.

### ***Answer***

I would prefer solar energy for running street lights, water pumps for irrigation and heating my house because of the following reasons:

1. Solar energy is a renewable source of energy which can be harnessed in most of the areas of the world.
2. It can be used for diverse purposes - to produce electricity in areas without access to the energy grid, to distill water in regions with limited clean water supplies and to power satellites in space.
3. Solar energy systems do not require a lot of maintenance. Most reliable solar panel manufacturers give 20-25 years warranty. Also, as there are no moving parts, there is no wear and tear. So, after covering the initial cost of the solar energy system, it requires little spending on maintenance and repair work.
4. It saves fossil fuels like coal and petroleum, used to produce electricity and also reduces electricity bills.

### **Question 2**

More people in the rural areas are now opting for LPG cylinders for domestic cooking. Do you think LPG should be replaced by biogas? Give reasons to support your answer.

***Answer***

Yes, I believe that LPG should be replaced by biogas because of the following reasons:

1. Renewable and sustainable — Biogas is produced from organic waste, making it a renewable energy source and reducing reliance on finite fossil fuels.
2. Effective waste management — Biogas production helps in the decomposition of organic waste, offering an efficient waste management solution and reducing landfill waste.
3. Lower emissions — Biogas emits fewer greenhouse gases and pollutants compared to LPG, leading to improved air quality and reduced environmental impact.
4. Cost-effectiveness — In areas with abundant biomass waste, biogas production can be a cost-effective alternative, reducing the reliance on costly LPG cylinders.
5. Energy independence — Biogas provides a decentralized energy source, promoting energy independence in rural areas by producing locally available biogas instead of relying on LPG cylinder refills.